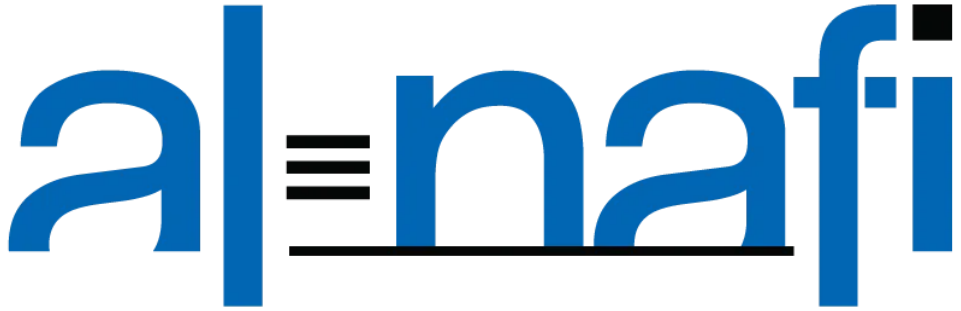
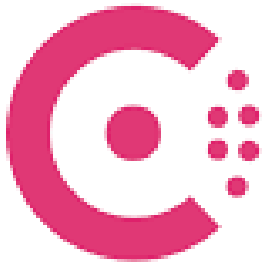


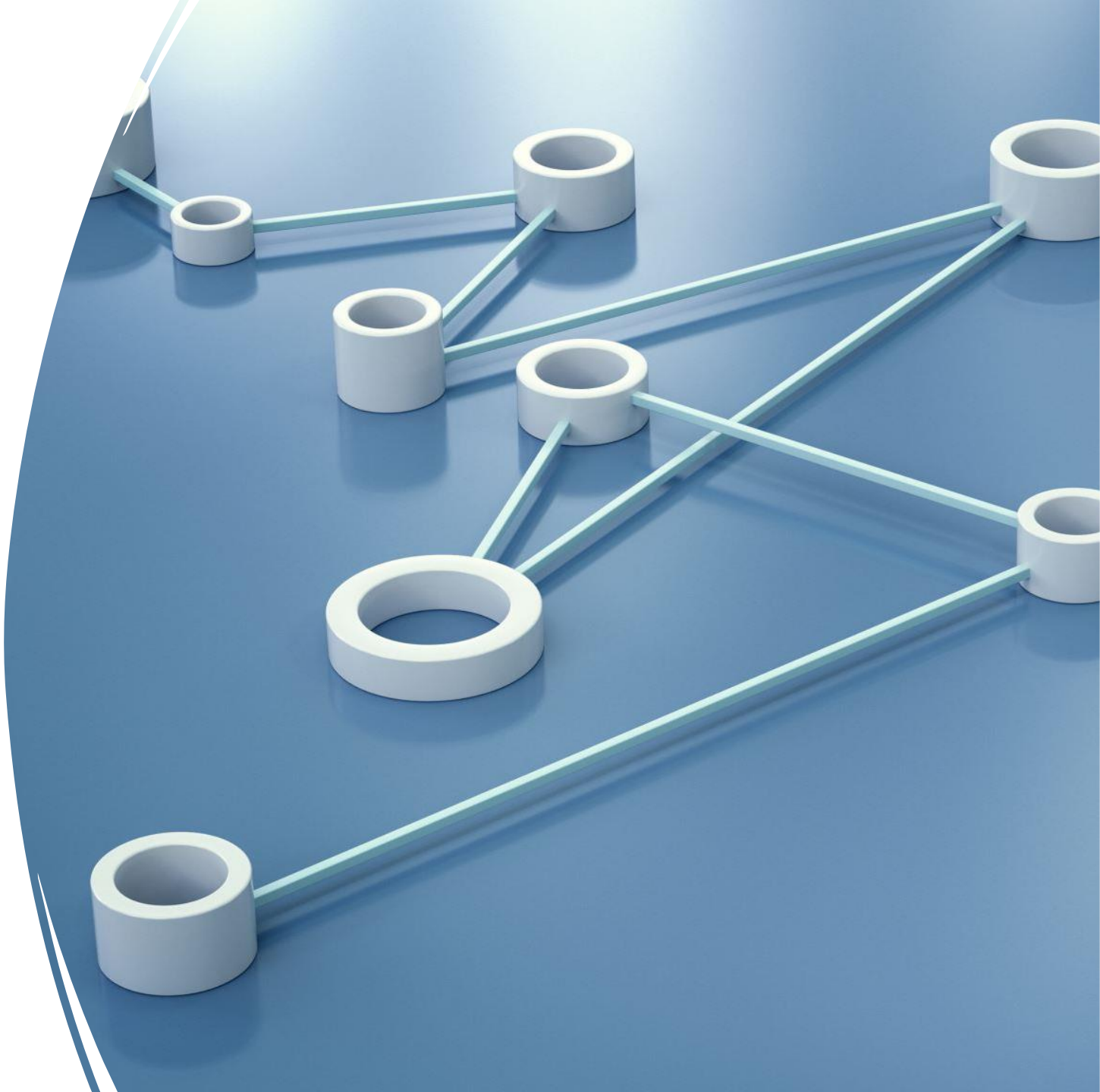


HashiCorp
Consul

What is Consul
Why Choose Consul
Core Concepts
Service Discovery
Service Mesh
Consul Architecture

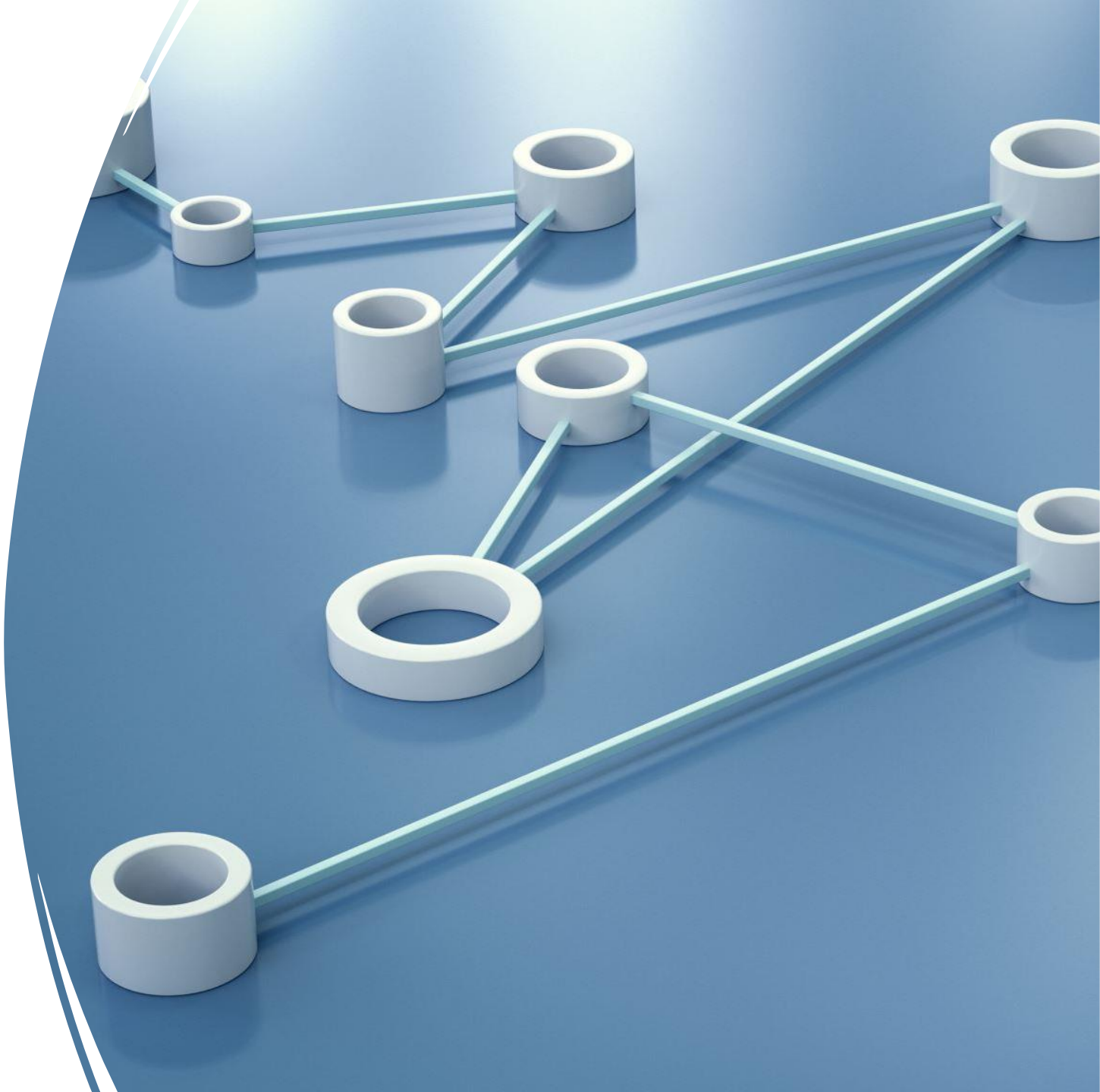
What is Consul

- HashiCorp Consul is a service networking solution that enables teams to manage secure network connectivity between services and across on-prem and multi-cloud environments and runtimes.



Cont...

- Consul Networking tool, service mesh solution providing full feature control of service discovery, configuration and segmentation challenges



Why Choose Consul (Service Meshes)

- Consul's service mesh allows organizations to securely connect and manage their network services across multiple different environments.
- Consul ensures that all service-to-service communication is authorized, authenticated, and encrypted.
- Consul includes traffic management capabilities like
  - load balancing and traffic splitting, which help developers perform
  - canary testing, A/B tests, and blue/green deployments.
  - Consul also includes health check and observability features.

Cont…

- Consul is platform agnostic —
  - it supports any runtime (Kubernetes, EKS, AKS, GKE, VMs, ECS, Lambda, Nomad) and
  - any cloud provider (AWS, Microsoft Azure, GCP, private clouds).
- This makes it one of the most flexible service discovery and service mesh platforms.
- While other service mesh software provides support for multiple runtimes for the data plane, they require you to run the control plane solely on Kubernetes. With Consul, you can run both the control plane and data plane in different runtimes.

DNS Tool

- Consul was originally designed as a centralized service registry for any cloud environment that dynamically tracks services as they are added, changed, or removed within a compute infrastructure. Consul maintains a catalog of these registered services and their attributes, such as IP addresses or service name.

Cont…

- As a result, Consul can also provide basic DNS functionality, including lookups, alternate domains, and access controls.
- Since Consul is platform agnostic, you can retrieve service information across both cloud and on-premises data centers.
- Consul does not natively support some advanced DNS capabilities, such as filters or advanced routing logic.

Configuration Management

- There are many configuration management tools, however, they typically focus on static provisioning.
- Consul enables you to dynamically configure your services based on service and node state. Both static and dynamic configuration are important and work well together.
- Since Consul offers a number of different capabilities, there are times when its functionality overlaps with other configuration management tools.

Cont…

- For example, Chef and Puppet are configuration management tools that can build service discovery mechanisms. However, they only support configuration information that is static. As a result, the time it takes to implement updates depends on the frequency of conversion runs (several minutes to hours). Additionally, these tools do not let you incorporate the system state in the configuration. This could lead to load balancers sending traffic to unhealthy nodes, further exacerbating issues. Supporting multiple datacenters is also challenging with these tools, since a central group of servers must manage all datacenters.
- Consul's service discovery layer is specifically designed to dynamically track and respond to your service's state. By using the integrated health checking, Consul can route traffic away from unhealthy nodes, allowing systems and services to gracefully recover. In addition, Consul’s service discovery layer works with Terraform. Consul-Terraform-Sync (CTS) automates updates to network infrastructure based on dynamic changes to each service. For example, as services scale up or down, CTS can trigger Terraform to update firewalls or load balancers to reflect the latest changes. Also, since each datacenter runs independently, supporting multiple datacenters is no different than supporting a single datacenter.

Cont…

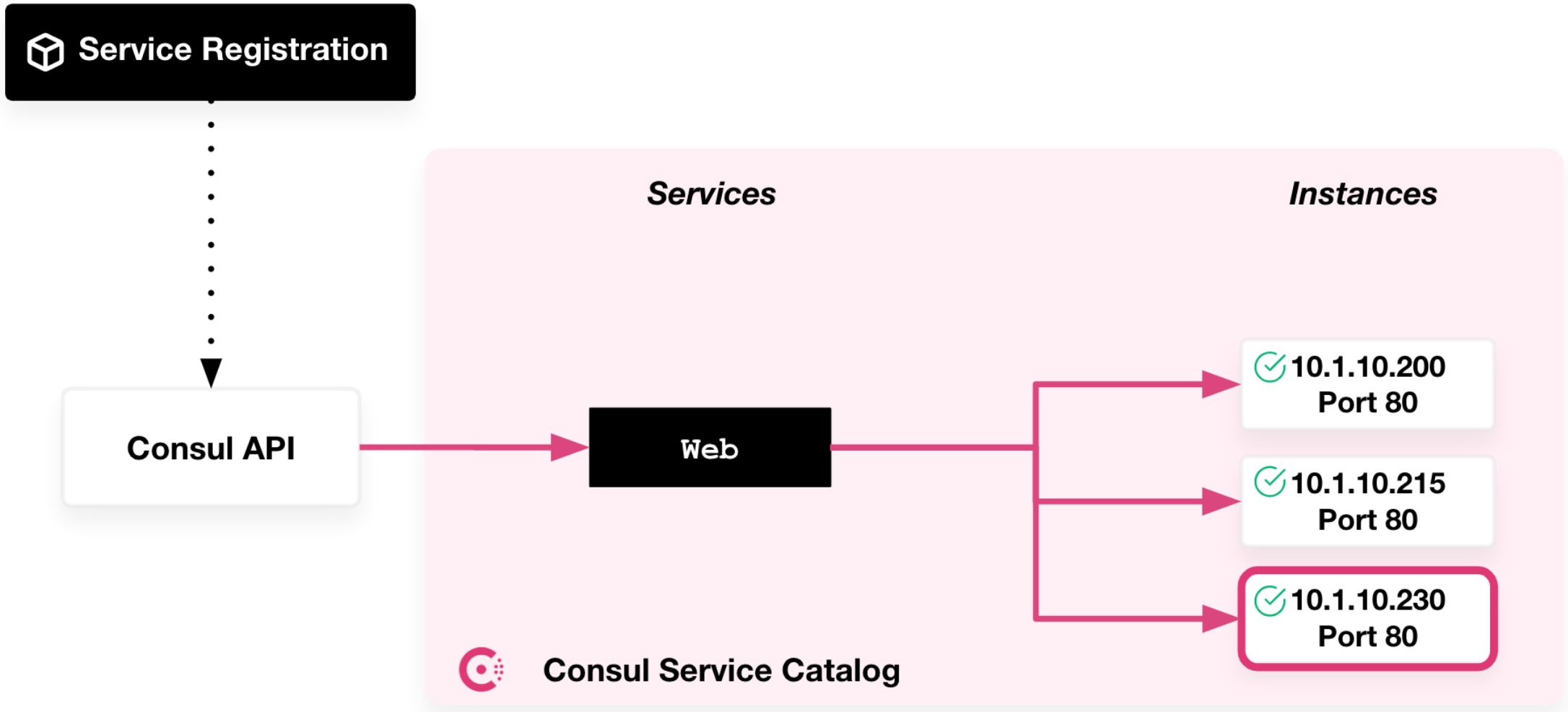
- Consul is not a replacement for other configuration management tools. These tools are still critical for setting up applications, including Consul.
- Static provisioning is best managed by existing tools, while Consul enables you to leverage dynamic configuration and service discovery.

API Gateways

- The Consul API Gateway is an implementation of the Kubernetes Gateway API. Traditionally, API gateways are used for two things:
- Client traffic management and
  - Consul service Mesh is responsible for routing inbound client requests to the mesh based on defined routes.
- API lifecycle management.
  - At this time, the Consul API Gateway does not support API lifecycle management.

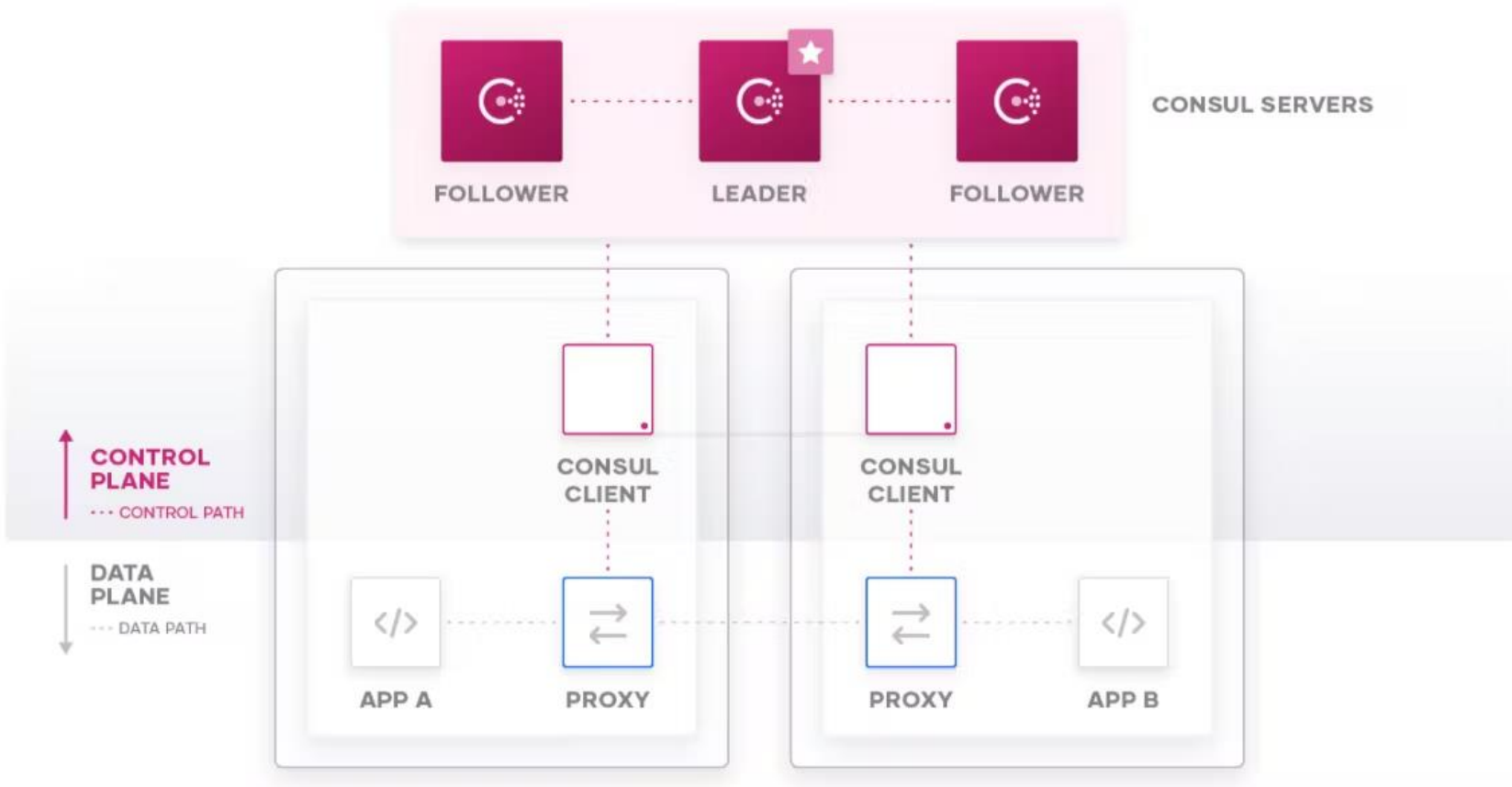
Service Discovery

- *Service discovery* helps you discover, track, and monitor the health of services within a network.
- Service discovery registers and maintains a record of all your services in a *service catalog*. This service catalog acts as a single source of truth that allows your services to query and communicate with each other.

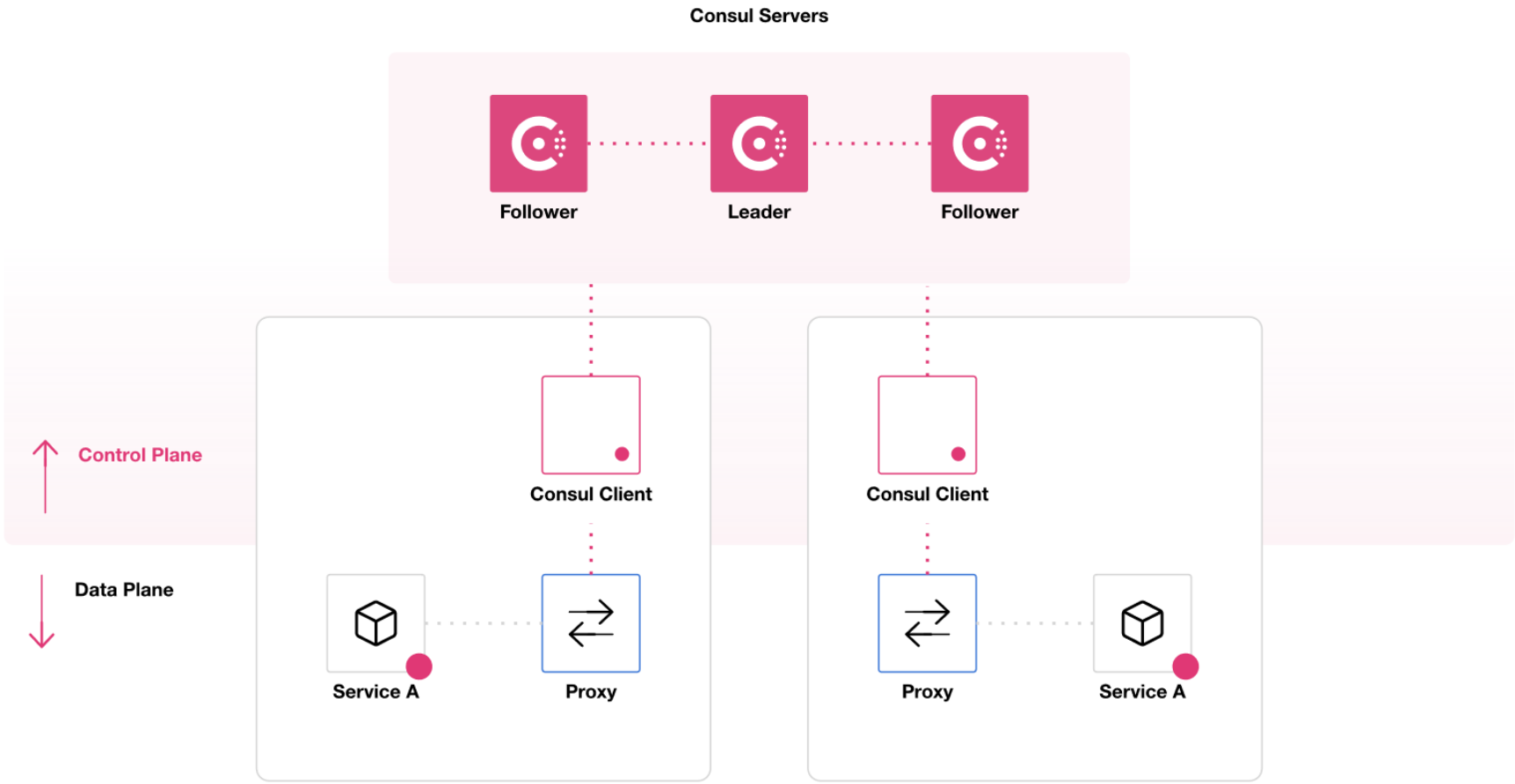

Service Registration
Consul API
Services
Instances
Web
10.1.10.200
Port 80
10.1.10.215
Port 80
10.1.10.230
Port 80
Consul Service Catalog

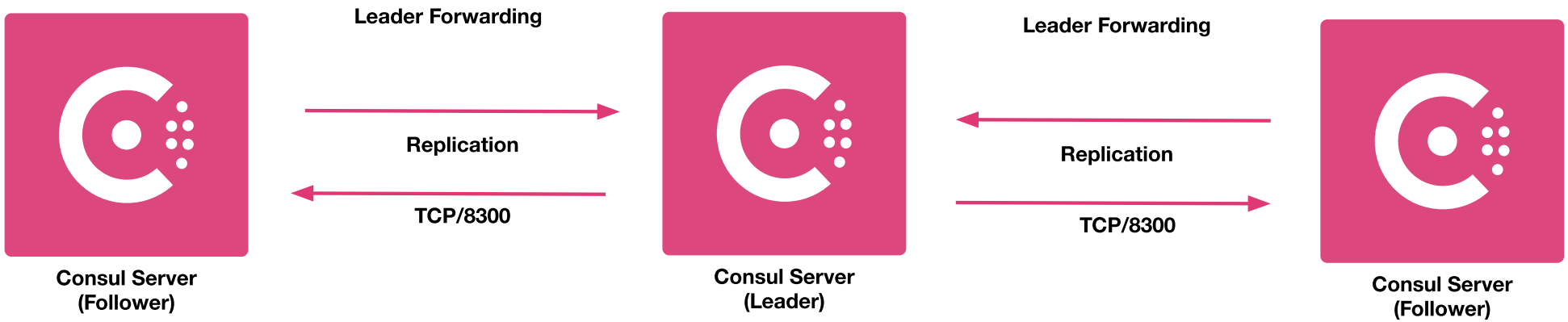
Service Mesh

- A *service mesh* is a dedicated network layer that provides secure service-to-service communication within and across infrastructure, including on-premises and cloud environments.


FOLLOWER
LEADER
FOLLOWER
CONSUL SERVERS
CONTROL PLANE
CONTROL PATH
DATA PLANE
DATA PATH
CONSUL CLIENT
CONSUL CLIENT
APP A
PROXY
PROXY
APP B

Consul Architecture




Leader Forwarding
Replication
TCP/8300
Consul Server
(Follower)
Consul Server
(Leader)
Leader Forwarding
Replication
TCP/8300
Consul Server
(Follower)

- **Datacenters**
- **Clusters**
- **Agents**
- **Server agents**
- **Client agents**